

John Matthew Gan

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Summary

Data & AI engineer with experience building cloud-based data, AI, and backend applications across data engineering, data lakehouses, LLM applications, document intelligence, and REST APIs. Hands-on with Python, SQL, AWS, Databricks, Apache Spark, and FastAPI in data transformations, addressing data quality issues, integrating AI capabilities, automating workflows, and production-oriented application development.

Skills

Languages: Python, SQL, JavaScript, TypeScript

Data & AI: Apache Spark, PySpark, Databricks, Pandas, ETL/ELT, Data Transformation, Data Quality, Data Modeling, Data Warehouses, Medallion Architecture, LLMs, RAG, NLP, TensorFlow, PyTorch, Computer Vision

Cloud & Software Engineering: AWS (Lambda, Bedrock, Athena, Glue, Lex, CloudWatch), Fast API, REST APIs, Node.js, Next.js, Gunicorn, Git, Jupyter, Agile Development, Unit Testing

Work Experience

Pointwest Technologies Corporation

June 2024 – present

AI / Data Engineer

- Built an Amazon Lex property search chatbot using AWS Lambda and S3 with conversational state handling, LLM-based data retrieval and validation, and open-ended user queries.
- Developed the LLM processing and backend for a recruiter-facing resume screening application supporting PDF and DOCX files, LLM output validation, and structured job fit assessment and bulk processing of job applications.
- Designed Bronze and Silver data mappings and automated SQL generation for two data lakehouse projects spanning over 22 tables across Databricks/PySpark/S3 and AWS Glue/Athena/S3. Defined schema relationships and data quality validation. Addressed invalid values, duplicates, nulls, and mismatches through normalization and layer-level testing across multiple environments.
- Rebuilt an ICD-10-CM medical coding copilot around human medical coding workflows using LLM-based concept extraction and disambiguation with deterministic Index lookup, XML traversal, and Tabular List validation.
- Extended a client-facing KYC document processing platform with new document templates and extraction validation. Investigated and implemented better data validation and error handling to improve processing reliability.
- Built reusable Python and Google Sheets-based data evaluation tools for automated batch testing and metric reporting across projects to replace manual evaluation workflows.

DLSU AdRIC - TE3D House

September 2023 – December 2023

Backend Developer

Manila, Metro Manila

- Built and deployed a FastAPI/Gunicorn REST backend on an Ubuntu VM to expose a low-level automatic speech recognition system to a web app. Resolved chatbot dialogue tree pathing issues in DialogFlow CX on Google Cloud.

Projects

LRT/MRT Train Car Calculator

November 2024 – August 2025

JavaScript, TypeScript, REST APIs, Cloudflare Workers, Bootstrap, SVG, LocalStorage

- Built and publicly deployed a commuter-facing web application and a Cloudflare Workers API server that determines the most optimal train cars based on station exits, travel direction, and train configuration across the LRT-1, LRT-2, and MRT-3.
- Performed data collection using Google Earth and on-site ocular inspections. Implemented server-side calculations and validation, flexible API input modes, persistent client-side settings, and dynamic SVG visualization of the recommended train cars.
- Iteratively improved the application through feature releases, bug fixes, UI improvements, and data updates.

Affect regression of tweets from the Philippines using emoji-based labels

September 2022 – August 2023

Python, Pandas, TensorFlow, PyTorch, CNN, BERT, NLP

- Trained CNN and BERT models to predict sentiment and emotion intensity in Philippine tweets with a large automatically annotated dataset and emoji lexicon that assigns continuous scores across emotions and sentiments.

Education

De La Salle University - Manila

2019 – 2023

Bachelor of Science in Computer Science with Specialization in Software Technology

Manila, Metro Manila